

Global Sorting, Morse Indices, and the Route-A Boundary of the Brockett Double-Bracket Flow

Route-A structural certificate C185

Abstract

For every real symmetric simple-spectrum orthogonal orbit and every strictly increasing diagonal target, we close the Brockett flow $\dot{H} = [H, [H, N]]$ without a dimension cutoff. The flow is globally isospectral and satisfies $d\text{Tr}(HN)/dt = \|[H, N]\|_F^2$. Its $n!$ equilibria are diagonal permutations.

Keywords: double bracket; isospectral flow; matrix sorting; Morse index; gradient dynamics.

中文摘要

本文对任意维实对称单谱正交轨道与严格递增对角目标，完整刻画 $\dot{H} = [H, [H, N]]$ ：解全局存在并保持谱，势函数导数恰为交换子的 Frobenius 范数平方；全部平衡点是 $n!$ 个对角排列。

关键词：双括号流；等谱流；矩阵排序；Morse 指标；梯度动力学。

1 Frozen family and global identities

Let

$$\lambda_1 < \cdots < \lambda_n, \quad \nu_1 < \cdots < \nu_n,$$

put $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $N = \text{diag}(\nu_1, \dots, \nu_n)$, and work on $\mathcal{O}_\Lambda = \{Q\Lambda Q^\top : Q \in O(n)\}$. The equation and its gradient sorting role are classical Brockett [1]; no priority is claimed here.

Set $K(H) = [H, N]$. It is skew-symmetric and $\dot{H} = [H, K(H)]$. If $\dot{Q} = -K(H)Q$, orthogonality is preserved and $H = Q\Lambda Q^\top$ solves the equation. Thus the compact orbit is invariant, all spectral traces are constant, and the smooth vector field is complete. Cyclicity of trace gives the strict identity

$$\frac{d}{dt} \text{Tr}(HN) = \text{Tr}([H, K]N) = -\text{Tr}(K^2) = \|K\|_F^2. \quad (1)$$

Consequently equilibrium is equivalent to $[H, N] = 0$. The strict target makes H diagonal, and the simple orbit spectrum makes its diagonal a permutation of the λ_i . There are exactly $n!$ equilibria $D_\pi = \text{diag}(\lambda_{\pi(1)}, \dots, \lambda_{\pi(n)})$.

Structure	Exact consequence	Boundary
skew Lax generator	global isospectral flow	not a fixed quantum operator
squared-norm derivative	strict gradient ordering	no nonconstant recurrence
simple/strict spectra	$n!$ isolated equilibria	target repetitions give zero modes

Revision-round focus. Round 0 freezes the all-size phase space, Lax proof, Lyapunov identity, and permutation equilibrium count.

Limitations and nonclaims. The paper does not claim novelty for the Brockett flow, a full degenerate critical-set or cell-closure theorem, arithmetic local data, an Euler factor, a root number, automorphy, a target divisor, a Hilbert–Polya operator, or Route-B authorization. The single source is an ownership lock, not an exhaustive literature review. Internal drafting checks are not external peer review.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

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References

- [1] R. W. Brockett, “Dynamical systems that sort lists, diagonalize matrices, and solve linear programming problems,” *Linear Algebra and its Applications* 146 (1991), 79–91. DOI: 10.1016/0024-3795(91)90021-N.